Calvin Deutschbein (they/them)

CONTACT INFORMATION

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ACADEMIC APPOINTMENTS

Assistant Professor of Computer Science, Willamette University — 2021 to present
School of Computing & Information Sciences, 2022-
College of Arts & Sciences, 2021-2022

Adjunct Professor, Elon University — 2020
Department of Computer Science

Instructor, The University of North Carolina at Chapel Hill — 2018
Department of Computer Science

Research Scholar, Semiconductor Research Corporation — 2018 to 2021
SRC Research Scholars Program

EDUCATION

The University of North Carolina at Chapel Hill, Chapel Hill, NC

Ph.D., Computer Science, August 2021

- Thesis: *Mining Secure Behavior of Hardware Designs*
- Advisor: Cynthia Sturton
- Area: Hardware Security

M.S., Computer Science, August 2017

- Thesis: *Multi-core Cyclic Executives for Safety-Critical Systems*
- Advisor: Sanjoy Baruah
- Area: Real-Time Systems

The University of Chicago, Chicago, IL

B.S., Computer Science, March 2015

- Thesis: *Performance and Energy Limits of a Processor-integrated FFT Accelerator*
- Advisor: Andrew A. Chien
- Area: Computer Architecture

B.A., Mathematics, March 2015

EXTERNAL RESEARCH FUNDING

National Science Foundation's Scholarships in Science, Technology, Engineering, and Mathematics — 2022 to 2028
- NSF Award # 2221694,
- Co-Principal Investigator 2024-, Senior Personnel 2022-2024
- Total Intended Award Amount: $1,499,246.00

Collaborative Research: SaTC: CORE: Medium: Hardware Security Insights: Analyzing Hardware Designs to Understand and Assess Security Weaknesses and Vulnerabilities — 2023 to 2027
- NSF Award # 2247756,
- Principal Investigator
- Total Intended Award Amount: $106,000.00

Refereed Journal Publications

[1] C. Deutschbein, A. Meza, F. Restuccia, R. Kastner, C. Sturton. Isadora: automated information-flow property generation for hardware security verification In: *Journal of Cryptographic Engineering*, November 2022.
doi:10.1007/s13389-022-00306-w

[2] C. Deutschbein, A. Meza, F. Restuccia, R. Kastner, C. Sturton. Toward Hardware Security Property Generation at Scale In: *IEEE Security & Privacy*, April 2022.
doi:10.1109/MSEC.2022.3155376

[3] R. Zhang, C. Deutschbein, P. Huang, C. Sturton. End-to-End Automated Exploit Generation for Processor Security Validation. *IEEE Design & Test Special Issue: Hardware Security Top Picks*. 2021.
doi:10.1109/MDAT.2021.3063314

[4] C. Deutschbein, T. Fleming, A. Burns, S. Baruah. Multi-core Cyclic Executives for Safety-Critical Systems. *Science of Computer Programming*, March 2019.
doi:10.1016/j.scico.2018.11.004

Refereed Conference Publications

[5] C. Deutschbein, O. Wyatt. Information Flow Paths from RTL Traces
In: *Interdisciplinary Conference on Electrics and Computer (INTCEC)*, September 2026.
Acceptance ratio: 45%
doi:10.48550/arXiv.2606.13860 *Forthcoming*

[6] C. Deutschbein, A. Stassinopoulos. "Test, Build, Deploy" – A CI/CD Framework for Open-Source Hardware Designs
In: *International Conference on Electrical, Computer and Energy Technologies (ICECET)*, April 2026.
Acceptance ratio: 37%
doi:10.1109/ICECET63943.2025.11472017

[7] C. Deutschbein, J. Ostler. RTL-Arrow: Hardware-to-Cloud Bridge
In: *Pacific Northwest Software Quality Conference*, October 2025.
doi:10.48550/arXiv.2606.13865

[8] C. Deutschbein, S. Ahmed Antu. Engaging with the Wiphala: A Code-Generation Hardened CS1 Final
In: *Consortium for Computing Sciences in Colleges - Rocky Mountain* October 2025.
doi:10.5555/3778141.3778149

[9] C. Deutschbein, J. Ostler, H, Raj. "vcd2df" – Leveraging Data Science Insights for Hardware Security Research
In: *International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, August 2025.
Acceptance ratio: 33%
doi:10.1109/ACDSA65407.2025.11165916

[10] S. Aftabjahani, M. Tehranipoor, F. Farahmandi, Farimah, B. Ahmed, R. Kastner, F. Restuccia, A. Meza, K. Ryan, N. Fern, J. van Woudenberg, R. Velegalati, C. Breunesse, C. Sturton, C. Deutschbein. Promising Directions for Automation of Security Assurance.
In: *Special Session: CAD for Hardware Security at 2023 IEEE 41st VLSI Test Symposium (VTS)*, June 2023.
Acceptance ratio: 28%
doi:10.1109/VTS56346.2023.10140100

[11] C. Deutschbein, A. Meza, F. Restuccia, R. Kastner, C. Sturton. Isadora: Automated Information Flow Property Generation for Hardware Designs.

In: *Proceedings of the 5th Workshop on Attacks and Solutions in Hardware Security (ASHES)*, November 2021.
Acceptance ratio: 31%
doi:10.1145/3474376.3487286

[12] C. Deutschbein, C. Sturton. Evaluating Security Specification Mining for a CISC Architecture.
In: *Proceedings of the IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, December 2020.
doi:10.1109/HOST45689.2020.9300291

[13] R. Zhang, C. Deutschbein, P. Huang, C. Sturton. End-to-End Automated Exploit Generation for Processor Security Validation.
In: *MICRO-51: Proceedings of the 51st Annual IEEE/ACM International Symposium on Microarchitecture*, October 2018.
doi:10.1109/MICRO.2018.00071

[14] C. Deutschbein, T. Fleming, A. Burns, S. Baruah. Multi-core Cyclic Executives for Safety-Critical Systems.
In: *Proceedings of the Third International Symposium on Dependable Software Engineering: Theories, Repositorys, and Applications, SETTA 2017*, October 2017.
doi:10.1016/j.scico.2018.11.004

[15] C. Deutschbein, S. Baruah. Preemptive Uniprocessor EDF Schedulability Analysis with Preemption Costs Considered.
In: *Proceedings of the 2016 IEEE Real-Time Systems Symposium (RTSS)*, November 2016.
doi:10.1109/RTSS.2016.047

[16] T. Thanh-Hoang, A. Shambayati, C. Deutschbein, H. Hoffmann, A. A. Chien Performance and energy limits of a processor-integrated FFT accelerator.
In: *Proceedings of the 2014 IEEE High Performance Extreme Computing Conference (HPEC)*, September 2014.
doi:10.1109/HPEC.2014.7040951

Invited Talks

[17] RTL-Arrow: Hardware-to-Cloud Bridge
Quality Jam, Pacific Northwest Software Quality Conference.
21 January, 2026.

[18] Leveraging Data Science Insights for Hardware Security Research
Reed College, Computer Science Colloqiuim.
12 November, 2025.

[19] Hardware Security Seminar
Portland State University, Computer Science Colloqiuim.
6 November, 2025.

[20] What is (Artificial) Intelligence?
Salem City Club.
14 March, 2025.

[21] "The Thinking Game"
Documentary Screening, Salem Cinema.
2 February, 2024.

[22] “Who ya gonna call?”: Cybersecurity for the Spectre Era.
Pacific University Mathematics, Computer Science, and Data Science Colloquium.
17 November, 2022.

[23] Isadora: Automated Information Flow Property Generation for Hardware Designs.
3rd Annual INTEL Side Channel Academic Program Workshop 2021.
11 November 2021.

[24] Isadora: Automated Information Flow Property Generation for Hardware Designs.
Workshop on Secure RISC-V Architecture Design (secrisc-v'21).
7 November 2021.

[25] Creating Information Flow Specifications.
Radix Presentation for Tortuga Logic.
20 August 2021.

[26] Extracting IF specifications from HW designs.
University of Illinois–Urbana Champaign
20 July, 2021.

[27] "Who ya gonna call?": Cybersecurity for the Spectre Era.
California State University Northridge Virtual Research Presentations: Computer Science and Cyber Security.
22 March, 2021.

Student Supervised Research Presentations

[28] S. Avery, L. Cochran (Equal Contribution). Streaming Feature Selection for Hardware Validation. In: *Formal Methods in Computer-Aided Design 2026*, September 2026. *Under submission.*

[29] O. Wyatt, Calvin Deutschbein. Constructing Information Flow Paths from Register Transfer Level Trace Data. In: *Consortium for Computing Sciences in Colleges - Northwest*, October 2025.

[30] J. DeYoung, Calvin Deutschbein. Aphrodite: Security Properties of RISC-V. In: *Northwest Scientific Association/American Association for the Advancement of Science - Pacific Division*, March 2023.

Teaching Materials

Continuous Integration and Continuous Delivery Security
- WGU MS-SWE project
- External Subject Matter Expert - Content Design & Assessment Design

Network Architecture and Advanced Cloud Computing
- WGU MS-SWE project
- External Subject Matter Expert - Content Design & Assessment Design

chiTCP - A simple, testable TCP stack
- The UChicago χ-Projects,
- Contributor
- 14 stars / 26 watching / 11 forks on GitHub

Chair Service

Poster Session, Consortium for Computing Sciences in Colleges Northwestern Region (CCSC-NW 2024).

Inquiry-Based Learning for Computing-Based Sciences, Title III Grant Quantitative Reasoning (QR) Summer Learning Circles

Poster Session, Consortium for Computing Sciences in Colleges Northwestern Region (CCSC-NW 2023).

Coding and Automation Session, Northwest Scientific Association-American Association for the Advancement of Science Pacific Divison 2023

Program Committee Service

Hardware and Architectural Support for Security and Privacy (HASP 2026), co-located with MICRO 2026. *Ongoing.*

Hardware and Architectural Support for Security and Privacy (HASP 2025), co-located with MICRO 2025. https://haspworkshop.org/2025/committee.html

Consortium for Computing Sciences in Colleges Northwestern Region (CCSC-NW 2024). https://www.ccsc.org/northwest/2024/committee.html

Hardware and Architectural Support for Security and Privacy (HASP 2024), co-located with MICRO 2024. https://haspworkshop.org/2024/committee.html

Consortium for Computing Sciences in Colleges Northwestern Region (CCSC-NW 2023). https://www.ccsc.org/northwest/2023/committee.html

Hardware and Architectural Support for Security and Privacy (HASP 2023), co-located with MICRO 2023. https://haspworkshop.org/2023/committee.html

Real-time And intelliGent Edge computing workshop (RAGE 2023), co-located with CPS-IoT Week 2023. https://rage-workshop.github.io/2023/organizers/

Hardware and Architectural Support for Security and Privacy (HASP 2022), co-located with MICRO 2022. https://haspworkshop.org/2022/committee.html

Sixth Workshop on Attacks and Solutions in Hardware Security (ASHES 2022), co-located with ACM CCS 2022. http://ashesworkshop.org/committees-2022